

MICHAEL S YAO	michaelsyao.com	hello@michaelsyao.com	Google Scholar	GitHub
OVERVIEW	Physician-scientist in training, building evidence-based ML systems for clinicians. Working on AI for clinical decision making, diagnostic reasoning, and clinical trials.			
EDUCATION	University of Pennsylvania, MD-PhD Bioengineering NIH F30 NRSA Fellow HHMI-NIBIB T32 Fellow Medical Education Concentration University of Pennsylvania, MS Computer Science Penn Engineering Teaching Award Recipient California Institute of Technology, BS Applied Physics Salutatorian Full-Ride Merit Scholarship Awardee			
EXPERIENCE	ML Scientist, Genentech San Francisco, CA 2025 Designed algorithms for optimizing personalized treatment strategies using LLMs, leveraging scientific literature, health record data, and insurance claim data to power predictions. AI Clinical Fellow, Glass Health Remote 2023-2024 Released and assessed clinical guideline articles as knowledge sources for large language models (LLM). Investigated applications of LLMs for medical education. PhD Researcher, Microsoft Research Redmond, WA 2022 Developed ML methods for accelerated MRI imaging. Proposed novel techniques for better generalization of MRI image reconstruction models. Software Engineer, Hyperfine Guilford, CT 2021 Implemented and validated algorithms for more robust MRI signal acquisition and image post-processing in MR software across 25+ hospital sites.			
SELECTED PUBLICATIONS	[1] Yao MS, Bastani O, Andersson A, Biancalani T, Bentaieb A, Iriondo C. Knowledgeable language models as black-box optimizers for personalized medicine. ICLR. (2025). Link [2] Yao MS, Chae A, Saraiya P, Kahn CE, Witschey WR, Gee JC, Sagreiya H, Bastani O. Evaluating acute image ordering for real-world patient cases via language alignment with radiological guidelines. Nature Communications Medicine. (2025). Link [3] Yao MS, Gee JC, Bastani O. Diversity by design: Leveraging distribution matching for offline model-based optimization. ICML. (2025). Link [4] Yao MS, Zeng Y, Bastani H, Gardner J, Gee JC, Bastani O. Generative adversarial model-based optimization via source critic regularization. NeurIPS. (2024). Link [5] Yang Y, Gandhi M, Wang Y, Wu Y, Yao MS, Callison-Burch C, Gee JC, Yatskar M. A textbook remedy for domain shifts: Knowledge priors for medical image analysis. NeurIPS (Spotlight). (2024). Link			
PROJECTS	Medical Education: Ethical Algorithms for the Modern Clinician Clerkship Connections C++ Programming: Advent of Code 2025 Project Euler Puzzles NRMP Match			